Diabetes mellitus

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- Diabetes mellitus: More commonly referred to as "diabetes" -- a chronic disease associated with abnormally high levels of the sugar glucose in the blood. Diabetes is due to one of two mechanisms:

- Inadequate production of insulin (which is made by the pancreas and lowers blood glucose), or Inadequate sensitivity of cells to the action of insulin.
- The two main types of diabetes correspond to these two mechanisms and are called insulin dependent (type 1) and non-insulin dependent (type 2) diabetes. In type 1 diabetes there is no insulin or not enough of it. In type 2 diabetes, there is generally enough insulin but the cells upon which it should act are not normally sensitive to its action.

- The signs and symptoms of both types of diabetes include increased urine output and decreased appetite as well as fatigue. Diabetes is diagnosed by blood glucose testing, the glucose tolerance test, and testing of the level of glycosylated hemoglobin (glycohemoglobin or hemoglobin A1C). The mode of treatment depends on the type of the diabetes.

- The major complications of diabetes include dangerously elevated blood sugar, abnormally low blood sugar due to diabetes medications, and disease of the blood vessels which can damage the eyes, kidneys, nerves, and heart.

ACUTE MANIFESTATIONS

- Polydipsia, polyuria, polyphagia, weight loss, DKA (type 1), hyperosmolar coma (type 2).
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CHRONIC COMPLICATIONS

- Small vessel disease retinopathy , glaucoma, neuropathy, nephropathy

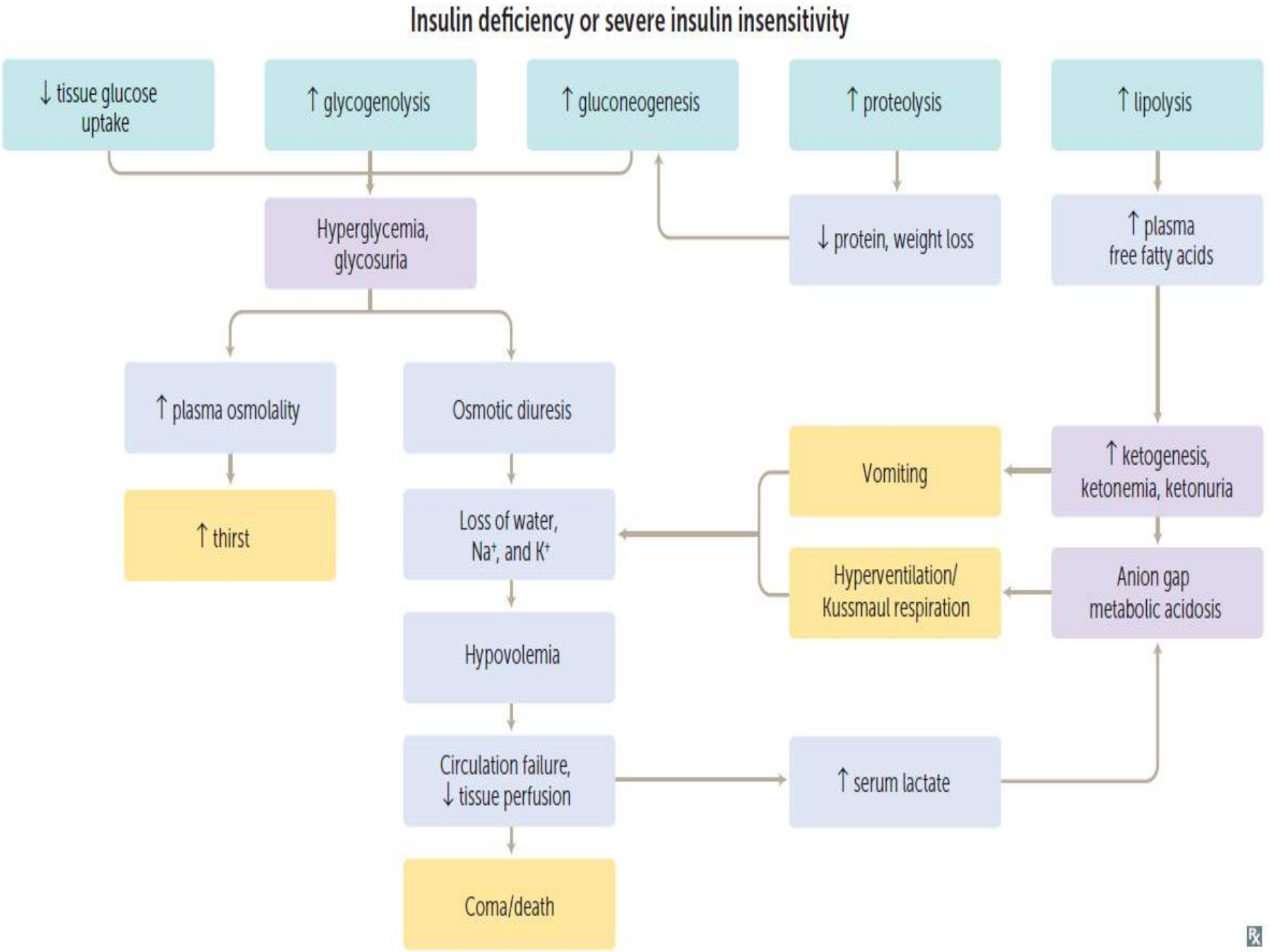
- proteinuria initially microalbuminuria and arteriolosclerosis, hypertension; both lead to chronic renal failure.

- Coronary artery diseases (MI) , STROKE , CATARACT

Large vessel atherosclerosis, CAD, peripheral vascular occlusive disease, gangrene , limb loss,

- cerebrovascular disease. MI most common cause of death.
- Osmotic damage sorbitol
- Neuropathy (motor, sensory and autonomic degeneration)
- Cataracts

TEST	DIAGNOSTIC CUT OFF	
HbA1c	≥ 6.5%	Reflects average blood glucose over prior 3 months
Fasting plasma glucose	≥ 126 mg/dL	Fasting for > 8 hours


Insulin deficiency or severe insulin insensitivity
↓ tissue glucose uptake
↑ glycogenolysis
↑ gluconeogenesis
↑ proteolysis
↑ lipolysis
Hyperglycemia, glycosuria
↓ protein, weight loss
↑ plasma free fatty acids
↑ plasma osmolality
Osmotic diuresis
↑ thirst
Loss of water, Na+, and K+
Vomiting
↑ ketogenesis, ketonemia, ketonuria
Hyperventilation/ Kussmaul respiration
Anion gap metabolic acidosis
Hypovolemia
Circulation failure, ↓ tissue perfusion
↑ serum lactate
Coma/death

Type 1 vs type 2 diabetes mellitus

Variable	Type 1	Type 2
1° DEFECT	Autoimmune destruction of β cells (eg, due to glutamic acid decarboxylase antibodies)	↑ resistance to insulin, progressive pancreatic β-cell failure
INSULIN NECESSARY IN TREATMENT	Always	Sometimes
AGE (EXCEPTIONS COMMONLY OCCUR)	< 30 yr	> 40 yr
ASSOCIATION WITH OBESITY	No	Yes
GENETIC PREDISPOSITION	Relatively weak (50% concordance in identical twins), polygenic	Relatively strong (90% concordance in identical twins), polygenic
ASSOCIATION WITH HLA SYSTEM	Yes (HLA-DR3 and -DR4)	No
GLUCOSE INTOLERANCE	Severe	Mild to moderate
INSULIN SENSITIVITY	High	Low
KETOACIDOSIS	Common	Rare
β-CELL NUMBERS IN THE ISLETS	↓	Variable (with amyloid deposits)
SERUM INSULIN LEVEL	↓	Variable
CLASSIC SYMPTOMS OF POLYURIA, POLYDIPSIA, POLYPHAGIA, WEIGHT LOSS	Common	Sometimes
HISTOLOGY	Islet leukocytic infiltrate	Islet amyloid polypeptide (IAPP) deposits

Diabetic ketoacidosis

- One of the most feared complications of diabetes. Usually due to insulin noncompliance or
- Increased insulin requirements from increased stress (eg, infection). Excess fat breakdown and ketogenesis from free fatty acids, which are then made into ketone bodies (β-hydroxybutyrate > acetoacetate).
- Usually occurs in type 1 diabetes, as endogenous insulin in type 2 diabetes usually prevents lipolysis.

SIGNS/SYMPTOMS

- DKA is Deadly: Delirium/psychosis, Kussmaul respirations (rapid/deep breathing), Abdominal Pain/nausea/vomiting, Dehydration. Fruity breath odor (due to exhaled acetone).

LABS

- Hyperglycemia, high H+, low HCO3
- – increased anion gap metabolic acidosis), increased blood ketone levels,
- leukocytosis. Hyperkalemia, but depleted intracellular K+ due to transcellular shift from lack of insulin and acidosis (therefore total body K+ is depleted).

TREATMENT

- IV fluids, IV insulin, and K+ (to replete intracellular stores); glucose if necessary to prevent hypoglycemia.

Renal threshold

- renal threshold is the concentration of a substance dissolved in the blood above which the kidneys begin to remove it into the urine.

- The plasma glucose concentration above which significant glucosuria occurs is called the renal threshold for glucose. Its value is variable, and deviations occur both above and below the commonly accepted "normal" threshold of **180 mg/dl.**

Detection of glucose in urine by Benedicts Qualitative method

- **Principal:**
- A reducing sugar tautomerises and forms enediols in an alkaline medium created by sodium carbonate. These enediols reduce copper sulphate to cuprous hydroxide on heating, which is converted into red PPT of cuprous oxide

Procedure :

- Take 5 ml of Benedicts Qualitative reagent in a test tube and add 8 to 10 drops of
- urine and then boil for 2 to 3 minutes

Observation

Color of PPT	Approx conc of carbohydrate (gm%)
GREEN	0.1-0.5 (+)
YELOW ORANGE	0.5-1.0(++)
Orange	1.0-1.5(+ + +)
Red	1.5-2.0(++++)
Brick red	ABOVE 2.0

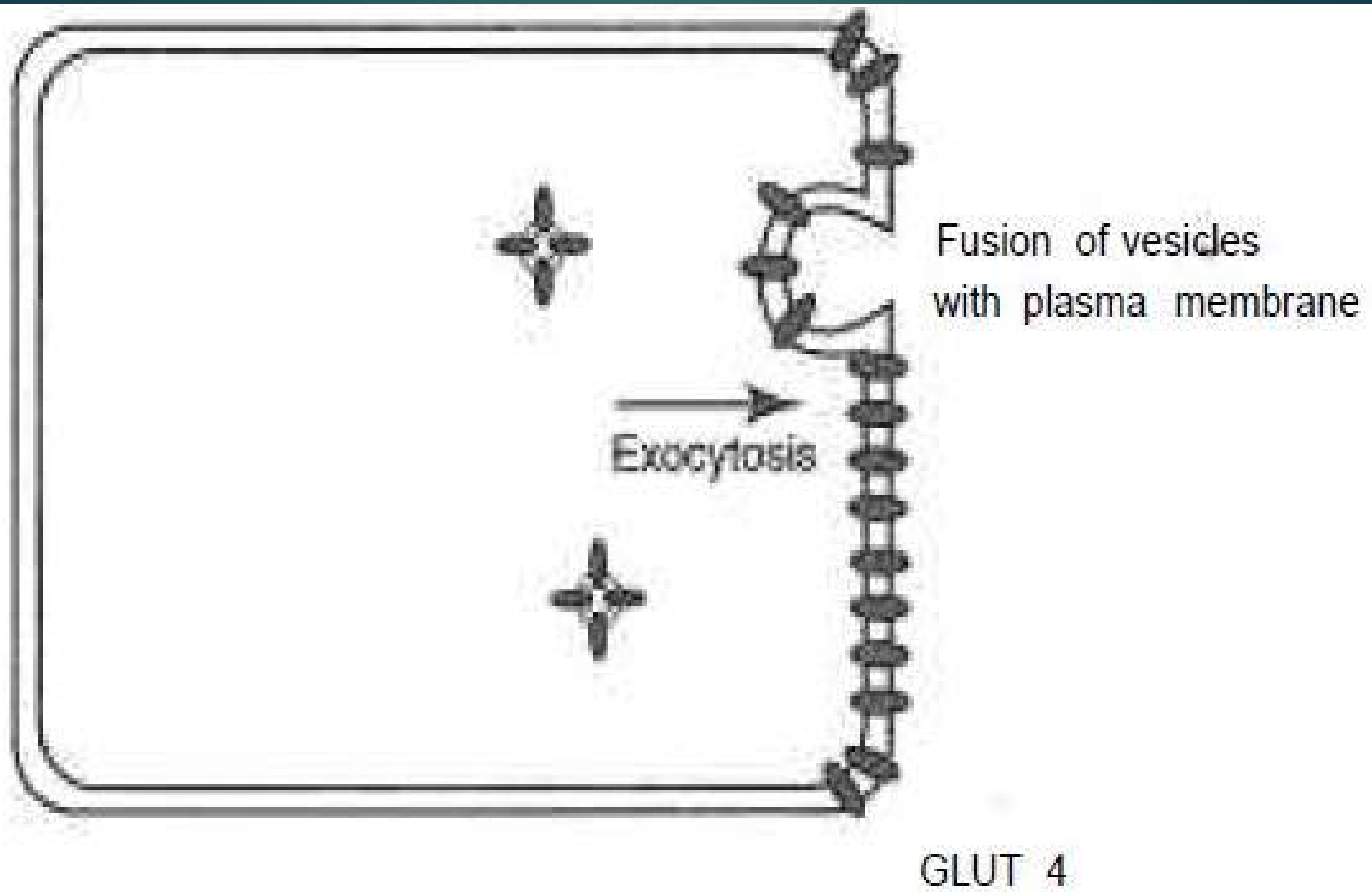
GLUCOSE TRANSPORT

- Glucose entry into most cells is concentration driven and independent of sodium. For glucose
- GLUT 1 and GLUT 3 mediate basal glucose uptake in most tissues, including brain nerves, and red blood cells. Their high affinities for glucose ensure glucose entry even during periods of relative hypoglycemia. At normal glucose concentration, GLUT 1 and GLUT 3 are at *Vmax'*

- GLUT 4 is in adipose tissue and muscle and responds to the glucose concentration in peripheral blood. The rate of glucose transport in these two tissues is increased by insulin, which stimulates the movement of additional GLUT 4 transporters to the membrane by a mechanism involving exocytosis

- Decreased insulin decreases the number of plasma membrane GLUT 4 transporters

Increased insulin increases the number of plasma membrane glucose transporters



Special function of GLUT 2

- Insulin secretion by the pancreatic ~-cells is biphasic. Glucose stimulates the first phase (within 15 minutes) with release of preformed insulin. The second phase (several hours) involves insulin synthesis at the gene level.

Name	Tissues	Km,Glucose	Functions
GLUT 1	Most tissues (brain, red cells)	~lmM	Basal uptake of glucose
GLUT 2	Liver Pancreatic ~-cells	-lSmM	Uptake and release of glucose by the liver ~-cell glucose sensor
GLUT 3	Most tissues	~lmM	Basal uptake
GLUT 4	Skeletal muscle Adipose tissue	-SmM	Insulin-stimulated glucose uptake

Normal blood glucose concentration is 4-6 mM (72-110 mg/dL).